

Timing	Lesson description	Additional materials	Comments
JuniorKoders Python Lesson 12. GUI Input 1			
Lesson objectives: Introduce students to GUI input and output of Python: Textbox, Radio Buttons, Spinner, using those components in a program.			
5	Last Lesson Review - Ask students about what they have learned during the last lesson. - Ask students if they still remember what they learnt last week.		
5	Introduction to Input & Output Elements in Python GUI Tell students they are going to learn about input and output in Python GUI. Ask if students know about some elements which can be used as input/output. Explain that some elements can be both input and output, even buttons. The teacher can show the main project, so the students have an idea of what they are going to learn today. Ask the students if they have studied HTML before because what we will be doing is similar to making an HTML Form. Summarise: the lesson today will be about input and output of Python GUI. It will be similar to creating a form because we have elements such as: textbox, radio buttons, and spinner	8. project.py	

10	Create Entry Widget Show & Explain • Show students the project and tell them that we want to learn how to make an entry/input field so users can give input to the program. • Show & Explain how to create an entry/input field in tkinter and how to get the data inputted in the entry field. Apply & Code: Create Entry Field • Teacher tell students to create a label, button and entry field in their program. Next, tell students to code the button so when it is clicked, the data inputted by the user in the entry field will be shown in the label. Save the project – ask students to save the project in a specific place with a proper name: StudentName_Projectname	1. Create input field.py	
10	Entry Widget Methods Show & Explain • Show students the project and tell them that other than getting the value inside the field, we also have other methods to configure the entry field such as focusing on a specific entry field and configuring the text inside the field. • Show & Explain to students the syntax used for those methods. Apply & Code: Entry Methods • Teacher tell students to practice using the methods explained previously by:	2. Input method.py	

	○ Creating two entry fields and two buttons where pressing the buttons will focus on the entry field listed on the button. ○ Creating another entry field and additional three buttons where when they are pressed, the text in the newly added entry field will be deleted, inserted with another text according to the position of the mouse in the entry field and appending text at the end. Save the project – ask students to save the project in a specific place with a proper name: StudentName_Projectname		
10	Create Radio Button Widget Show & Explain: Radio Button Widget ● Tell students about radio button widget. Tell them we use a radio button to limit the user's answer to only one from a list of possible answers. We can also call the widget option button. ● Show & explain to students how to make a radio button using the tkinter library. Apply & Code: Create Radio Button ● Teacher tell students to make 2 radio buttons in their program and see the result of their code. Save the project – ask students to save the project in a specific place with a proper name: StudentName_Projectname	3. Radio button.py	Until “Getting Radio Button Value” section, the teacher can just do one project and keep adding features to it for radio button widget.

5	Radio Button Value Show & Explain: Radio Button with Value • Ask students about their radio button they have created. Can they select them or they already selected and cannot unselect ?. tell students that it is okay because we have not completed the radio button yet. • Show & Explain to students how to differentiate between radio buttons by adding value to each radio button. Apply & Code: Radio Button Value • Teacher tell students to add value to the radio button they already made previously. Save the project – ask students to save the project in a specific place with a proper name: StudentName_Projectname	4.Radio Button with Value.py	
5	Categorising Radio Button Show & Explain: Radio Button Category • Tell students that by adding values, we can already differentiate between radio buttons so we can only select one. Ask students what if we have two sets of radio buttons in the program and by adding value, does it already differentiate between those two sets? the answer is no • Show & Explain to students how to categorised radio buttons by using the tkinter variable • Explain to students about the tkinter variable, which is a variable explicitly used for tkinter widget. The teacher explains that we are using the tkinter variable because the tkinter widget is used. The teacher also tells students what data types we can use for the tkinter variable.	5. Categorized Radio Button.py	List of Data Types can be used for tkinter variable: • Integer => <code>tkinter.IntegerVar()</code> • String => <code>tkinter.StringVar()</code> • Boolean => <code>tkinter.BooleanVar()</code> For More Information regarding the variable: https://www.python-course.eu/tkinter_variable_classes.php

	Apply & Code: Categorizing Radio Button Teacher tell students to add two more radio buttons and create a tkinter variable with string datatype to their project. Next, tell them to use the tkinter variable on the new two radio buttons they already added. Save the project – ask students to save the project in a specific place with a proper name: StudentName_Projectname		
5	Get Radio Button Value Show & Explain: Get Value from Radio Button Ask students if they see the difference of using the tkinter variable on the radio button widget. Tell students that there is another reason why we use tkinter variable on the radio button widget other than categorised the radio buttons in the program which is to get the value of radio button chosen Show & Explain to students how to get the value of the chosen radio button by using the tkinter variable assigned to it. Apply & Code: Getting Radio Button Value - Teacher tell students to make a function that will print the value of the tkinter variable in their program. - Tell students to assign the function to the radio button that has already been assigned a tkinter variable. - Tell students to check the shell to see the result of their code.	6. Getting Radio Button Value.py	

	Save the project – ask students to save the project in a specific place with a proper name: StudentName_Projectname		
10	Create Spinbox Widget Show & Explain: Spinbox Widget • Tell students about the Spinbox/Spinner widget that can be used for giving numeric input. • Show & explain to students how to create a spinbox widget and how to configure the number input of the widget. Teacher tell students that there is another way to configure the number input of the spinbox and explain to them how to do it. • Explain to students that we can also configure the states of the spinbox to change the availability of the widget. • Show & Explain to students how to get the value of a spinbox widget. Apply & Code: Create Spinner • Teacher tell students to practice by creating 2 spinbox in their program and tell them to configure the number in the widget using two methods of the configuration already explained to them. Save the project – ask students to save the project in a specific place with a proper name: StudentName_Projectname	7. Spinner.py	Spinbox states option: • Normal • Readonly • Disabled

20	Project - Sign Up Form Show & Explain: Sign Up Form • Show & Explain to students what is the project about and tell them the program should have: ○ Input field for inputting firstname, lastname, username, password, and email ○ radio buttons to choose a gender ○ spinbox to choose the number for the duration ○ a button that when pressed, will show all the data inputted by the user on the field provided. Individual Work: Creating Sign Up Form • Teacher tell students to create a similar project to what has been shown to them. • Remind students to add a validation where if one of the fields is left empty, the result of the button pressed will ask the user to input again. Save the project – ask students to save the project in a specific place with a proper name: StudentName_Projectname	8. Project.py	The teacher can help students to add a configuration on the entry widget where we can add “ show = ”*” ” to make all the text changed by the * character.
5	Last Overview Summing Up • We sum up by asking questions making sure the students remember the essential things. • We thank students for doing a great job overall.		